# Minimal idempotent subspaces of every dimension in evolution algebras

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**Abstract**

We answer the question in Andres Barei's Remark 3.5. For every integer $d \geq 2$ and every field of characteristic different from two, we construct an evolution algebra containing a minimal nonzero idempotent subspace of dimension $d$. The case $d = 3$ is realized explicitly in dimension five and its ambient minimality is checked in Lean 4. The construction has intrinsic embedding dimension exactly $2d-1$. We also record a direct criterion in dimension three and the changes over $\mathbb{Q}$, $\mathbb{R}$, and fields of characteristic two.

## 1. The five-dimensional answer to Remark 3.5

Let $E$ have natural basis $e_1, \ldots, e_5$, with $e_i e_j = 0$ for $i \neq j$, and put

$$u = e_1 + e_2 + e_3 + e_4 + e_5, \qquad v = e_2 - e_3, \qquad w = e_4 - e_5.$$

Define

$$e_1^2 = 4u + 2w, \quad e_2^2 = v, \quad e_3^2 = -v, \quad e_4^2 = v + w, \quad e_5^2 = -v - w.$$

The row-square matrix is

$$\begin{pmatrix} 4 & 4 & 4 & 6 & 2 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & -1 \\ 0 & -1 & 1 & -1 & 1 \end{pmatrix}.$$

The vectors $u, v, w$ are independent and direct multiplication gives

$$u^2 = 4u + 2w, \qquad uv = 2v, \qquad uw = 2v + 2w, \qquad v^2 = vw = w^2 = 0. \tag{1}$$

Thus $V = \mathrm{Span}\{u, v, w\} = E^2$ and (1) gives $V^2 = V$.

**Proposition 1.** *$V$ is a minimal nonzero idempotent subspace of $E$.*

*Hand-checkable proof.* Every proper subalgebra of $V$ lies in $W = \mathrm{Span}\{v, w\}$ and has zero multiplication. Indeed, if $x = au + bv + cw$ belongs to a subalgebra $U$ with $a \neq 0$, then

$$s = x^2 - 4ax = 4acv + 2a^2 w \in U, \qquad xs - 2as = 4a^3 v \in U.$$

Since $2a \neq 0$, this gives $v \in U$, then $w \in U$ from the first formula, and finally $u \in U$ from $x$. Hence $U = V$. If no such $x$ exists, then $U \subseteq W$ and $U^2 = 0$. An idempotent subspace is a subalgebra, so a nonzero idempotent subspace of $V$ must be all of $V$. □

The same equations show that $E$ has no nonzero idempotent element. For $x = au + bv + cw$, the $u$ and $w$ coordinates of $x^2 = x$ are $4a^2 = a$ and $2a^2 + 4ac = c$. If $a = 0$, then $x^2 = 0$ and $x = 0$; if $a \neq 0$, then $a = 1/4$ and the second equation reads $1/8 + c = c$. The realization and the ambient minimality assertion are kernel-checked in `Extension.lean`; the exact Gröbner computation in `exact_search.py` independently gives the zero-only idempotent solution. The Lean proof covers the $d = 3$ realization over $\mathbb{C}$; the general family below is proved on paper and checked exactly for $d = 2, 3, 4, 5$.

## 2. A family in every dimension and the bound $2d - 1$

Fix $d \geq 2$, put $m = d-1$, and assume char $\neq 2$. Let $E_d$ have natural basis $f_0, f_1^+, f_1^-, \ldots, f_m^+, f_m^-$, and set

$$u = f_0 + \sum_{i=1}^{m} (f_i^+ + f_i^-), \qquad w_i = f_i^+ - f_i^-, \qquad w_0 = 0,$$

with

$$f_0^2 = 4u + 2w_m, \qquad (f_i^+)^2 = w_i + w_{i-1}, \qquad (f_i^-)^2 = -w_i - w_{i-1}.$$

For $V_d = \mathrm{Span}\{u, w_1, \ldots, w_m\}$ and $W = \mathrm{Span}\{w_1, \ldots, w_m\}$ one computes

$$u^2 = 4u + 2w_m, \qquad uw_i = 2w_i + 2w_{i-1}, \qquad W^2 = 0. \tag{2}$$

The operator $Nw_i = w_{i-1}$ is nilpotent, and $L_u|_W = 2(I + N)$ is invertible. Hence $V_d^2 = V_d$ and the natural-basis squares span $V_d$, so $E_d^{(2)} = E_d^{(\infty)} = V_d \neq 0$.

If a subalgebra $U \subseteq V_d$ contains $u + z$ with $z \in W$, then $S = U \cap W$ is $N$-invariant and

$$(u + z)^2 - 4(u + z) = 2w_m + 4Nz \in S.$$

This vector is cyclic for the single Jordan block $N$: its successive images have leading coefficients 2 in positions $m, m - 1, \ldots, 1$. Thus $S = W$ and $U = V_d$. All other subalgebras lie in $W$ and square to zero. Therefore $V_d$ is minimal idempotent of dimension $d$.

This intrinsic algebra has optimal evolution-algebra embedding dimension $2d - 1$. For an embedding in an $n$-dimensional evolution algebra with column-square matrix $M$, $\operatorname{rank} M \geq d$ because $V_d^2 = V_d$. The embedded square-zero space $W$ has dimension $m$; the vectors $z \odot z$ span a subspace of $\ker M$ of dimension at least $m$ (choose $m$ independent coordinate projections on $W$). Hence

$$n = \operatorname{rank} M + \dim \ker M \geq d + m = 2d - 1.$$

The displayed construction attains the bound. This is an optimality theorem for this intrinsic family, not a claim that every three-dimensional minimal subspace needs five ambient dimensions.

## 3. Dimension three and the ground field

For a three-dimensional evolution algebra let $M$ be the column-square matrix and $W = \operatorname{im} M = E^2$. Over $\mathbb{C}$, the following are equivalent:

1. $E$ is nonsolvable and has no nonzero idempotent;
2. $\operatorname{rank} M = 2$ and a basis $u, v$ of $W$ satisfies $v^2 = 0$, $u^2 = au + bv$, $uv = (a/2)v$, with $a, b \neq 0$;
3. $W$ is isomorphic to $p^2 = p + q$, $pq = q/2$, $q^2 = 0$.

The proof reduces to the two-dimensional perfect algebra $W$. Writing $u^2 = au+bv$, $uv = cu+dv$, $v^2 = 0$, an idempotent with nonzero $u$-coordinate would, when $c \neq 0$, give a root over $\mathbb{C}$ of $2(bc-ad)x^2+(2d+a)x-1=0$. Thus $c=0$; perfectness gives $ad \neq 0$ and idempotent-freeness forces $2d = a$ and $b \neq 0$. This yields the criterion.

The field matters. Over $\mathbb{Q}$, the two-dimensional algebra $e_1^2 = e_2$, $e_2^2 = 2e_1$ is nonsolvable and idempotent-free over $\mathbb{Q}$: an idempotent would satisfy $x = 2y^2$, $y = x^2$, hence $2x^3 = 1$, excluded by the rational-root theorem. Over $\mathbb{R}$, every nonsolvable algebra of dimension at most two has a nonzero idempotent (in dimension two this follows from a real root of the odd cubic $ct^3 - dt^2 + at - b$), while the five-dimensional example above supplies a three-dimensional stable minimal subspace.

In characteristic two, Barei's original matrix becomes solvable, but the following different algebra works over every field of characteristic two:

$$e_1^2 = e_2 + e_3, \qquad e_2^2 = e_2 + e_3, \qquad e_3^2 = e_1 + e_3.$$

With $u = e_2 + e_3$ and $v = e_1 + e_2$, one has $E^2 = \mathrm{Span}\{u, v\}$ and $u^2 = v$, $uv = u$, $v^2 = 0$. Since $(au+bv)^2 = a^2 v$, the plane has no nonzero idempotent, while it squares to itself. This is certified in `CharacteristicTwo.lean`; the exact idempotent ideal is $(x_0, x_1, x_2)$.

## 4. Verification, credit, and open questions

The $d = 3$ case is checked in Lean 4 at the fixed public revision BareiRemarkThreeFive.lean. The characteristic-two witness is also checked in the local Lean file `CharacteristicTwo.lean`; the general-$d$ family, the dimension-three criterion, and the remaining field arguments are written proofs supplemented by exact symbolic computations. The note answers Barei's Remark 3.5 (with Muse Spark), and credits Hu–Wen, arXiv:2609.25023, Costoya–Fernández Ouaridi–Viruel, arXiv:2609.32784, and the classification of three-dimensional evolution algebras by Cabrera Casado–Siles Molina–Velasco. The literature check was bounded; no priority claim is made for the Jordan family or the characteristic- two example.

Open questions remain: can a different three-dimensional minimal idempotent subspace occur in a four-dimensional complex evolution algebra; how should minimal idempotent subspaces of dimension at least three be classified up to intrinsic isomorphism; and what is the prior-literature status of the exact families above?

**AI use.** The results were obtained with AI assistance in exploration, formalization, exact computation, and drafting. The stated Lean theorems, symbolic computations, and hand proofs were checked against the supplied source files; responsibility for the mathematical claims remains with the authors.